Experiment - 2

TestDisk: Open-Source Data Recovery Tool

recover a missing partition and repair a corrupted one using test disk

Installation

Linux : sudo apt-get install testdisk

macOS : brew install testdisk

Windows: Download the executable from the official website.

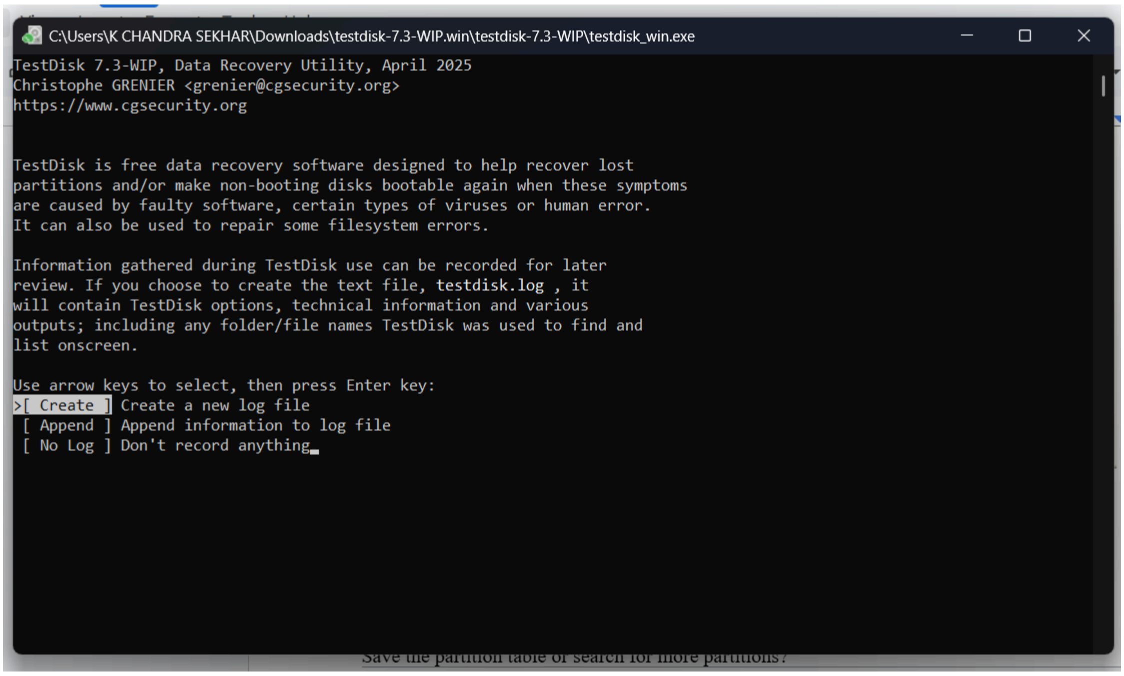
Link : https://www.cgsecurity.org/wiki/TestDisk_Download

NOTE:- if you using in windows please extract the .zip file first

Procedure

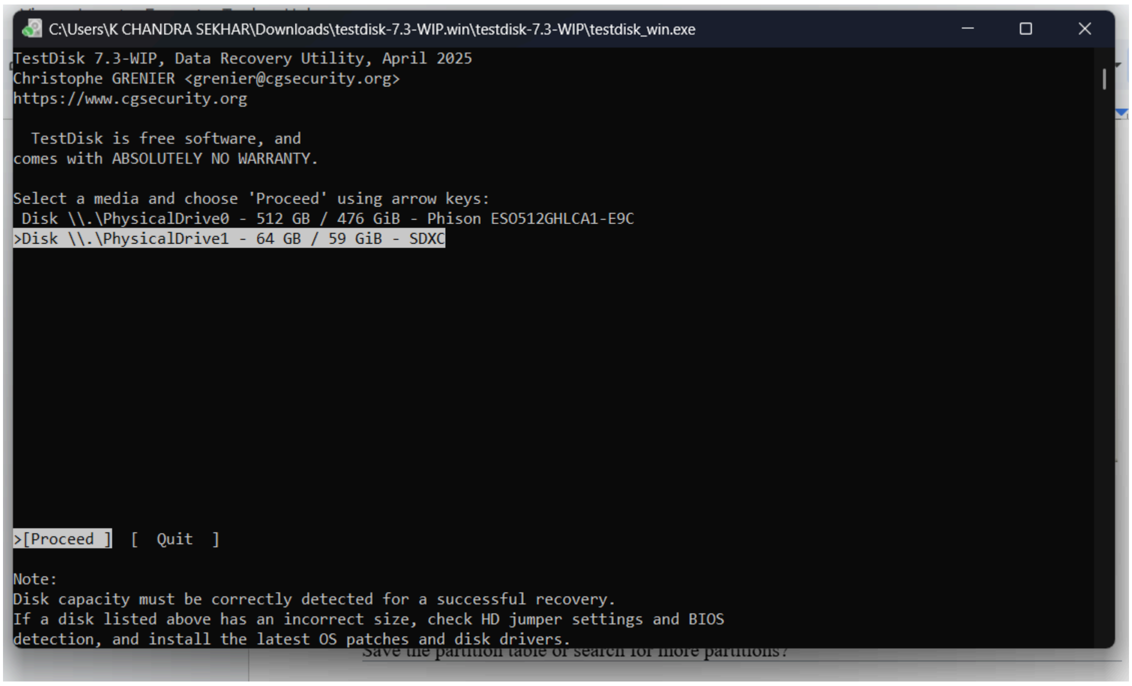
1. Create a Log File

- Launch TestDisk from your terminal or command prompt using sudo testdisk (or testdisk_win.exe on Windows).
- Select the [Create] option to generate a log file of the recovery session. This is helpful for future reference or troubleshooting.

- 

2. Select the Drive to Analyze

- TestDisk will display a list of all detected storage devices.
- Use the Up/Down arrow keys to highlight the drive that contains your lost data.
- Select [Proceed] to move to the next step.



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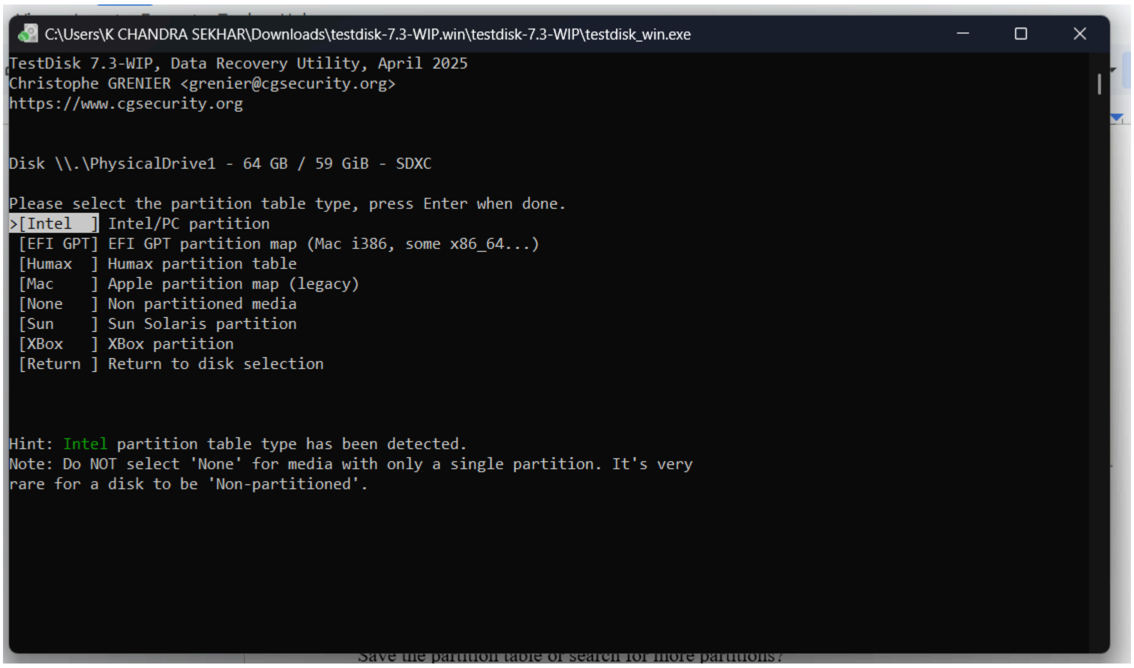
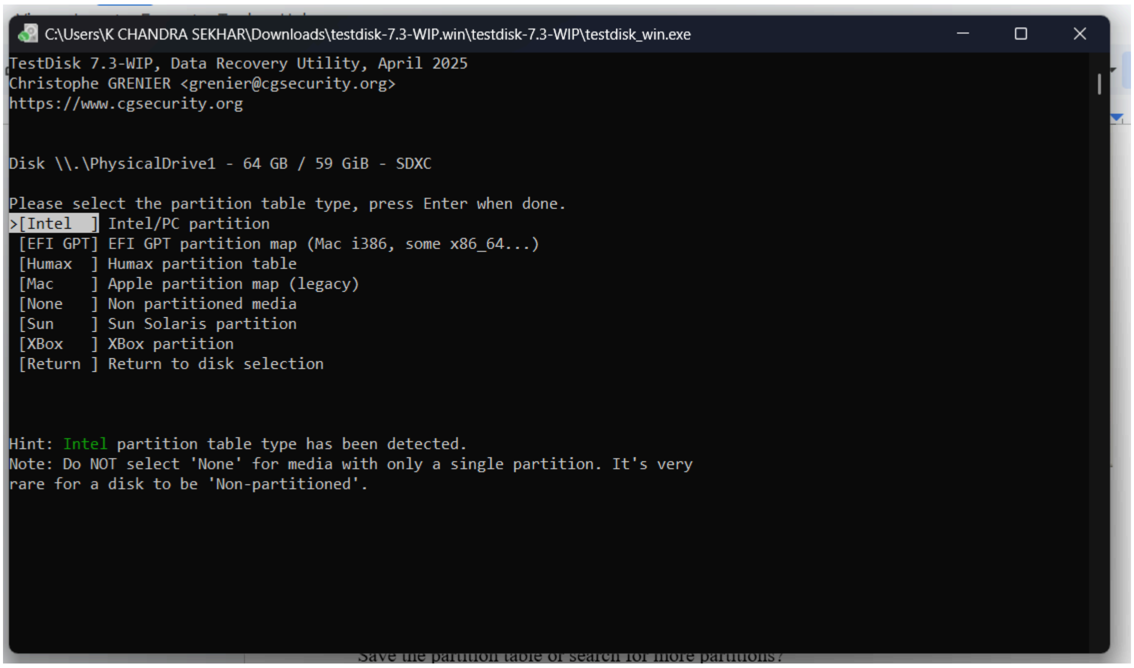
3. Choose the Partition Table Type

- TestDisk will automatically suggest the most likely partition table type (e.g., Intel/PC, EFI GPT).
- The default value is usually correct. Confirm the selection by pressing Enter.



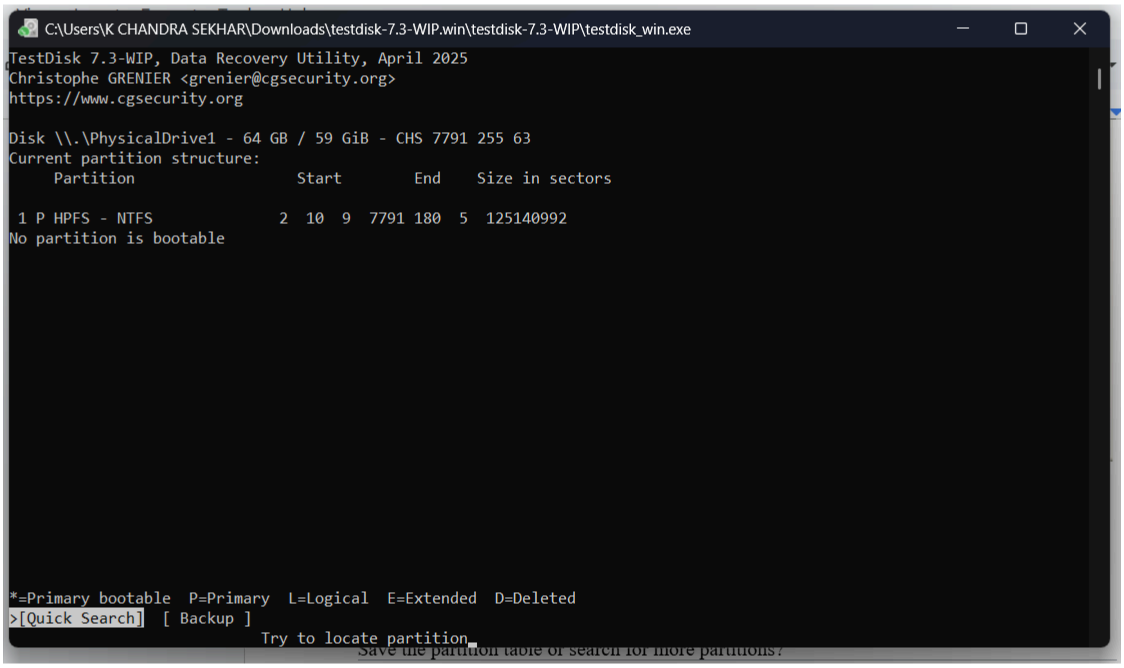
4. Analyze Current Partition Structure

- From the main menu, choose [Analyse] and press Enter.
- This will display the current partition table and check for errors or missing partitions.

- 

5. Perform a Quick Search

- After the analysis, you will be prompted to perform a [Quick Search].
- Select it and press Enter to scan the drive for lost partitions.



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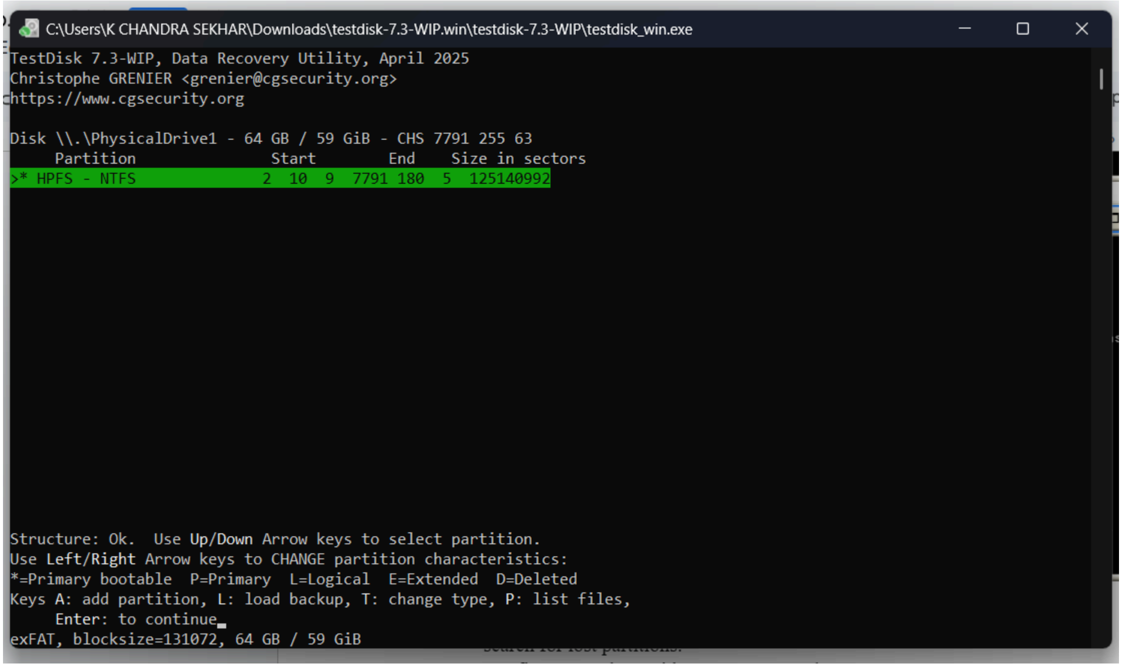
- Once a partition is found, you can press p to list its files. Deleted files and folders often appear in red.
- Press q to return to the search results.

6. Perform a Deeper Search

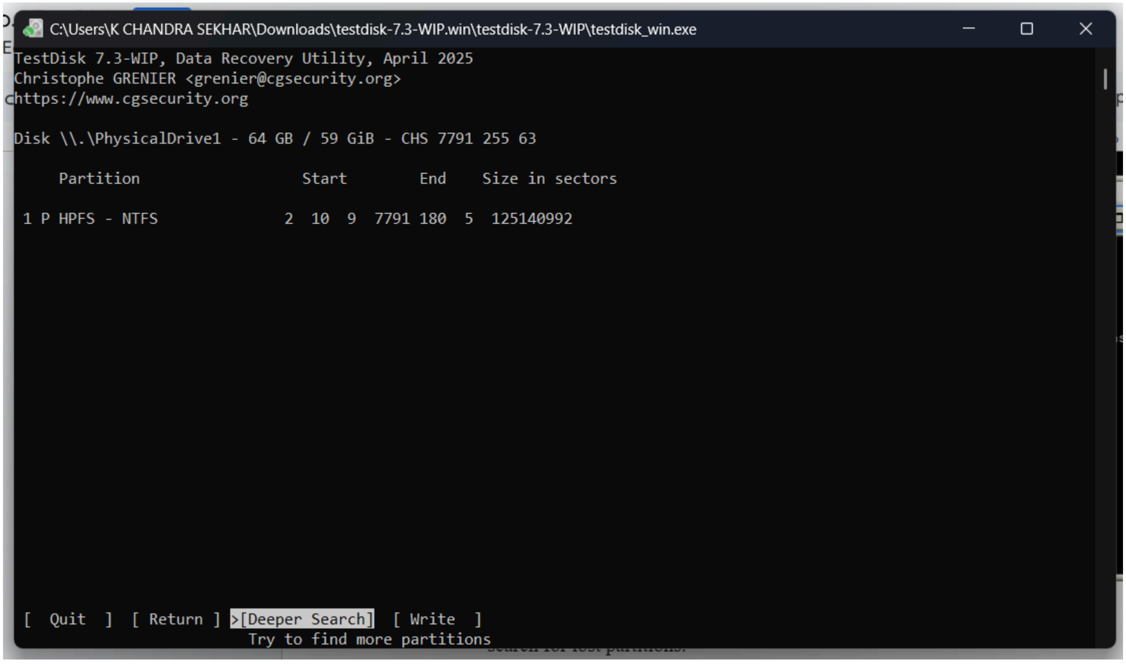
- If the Quick Search fails to find your lost partitions, select [Deeper Search]
- This process can take a long time, as it scans the entire drive, block by block, to find remnants of partition structures.



- Again, use p to preview files and confirm if a found partition is the one you are looking for.

7. Modify Partition Status

- After finding the correct partitions, use the Left/Right arrow keys to set their status.

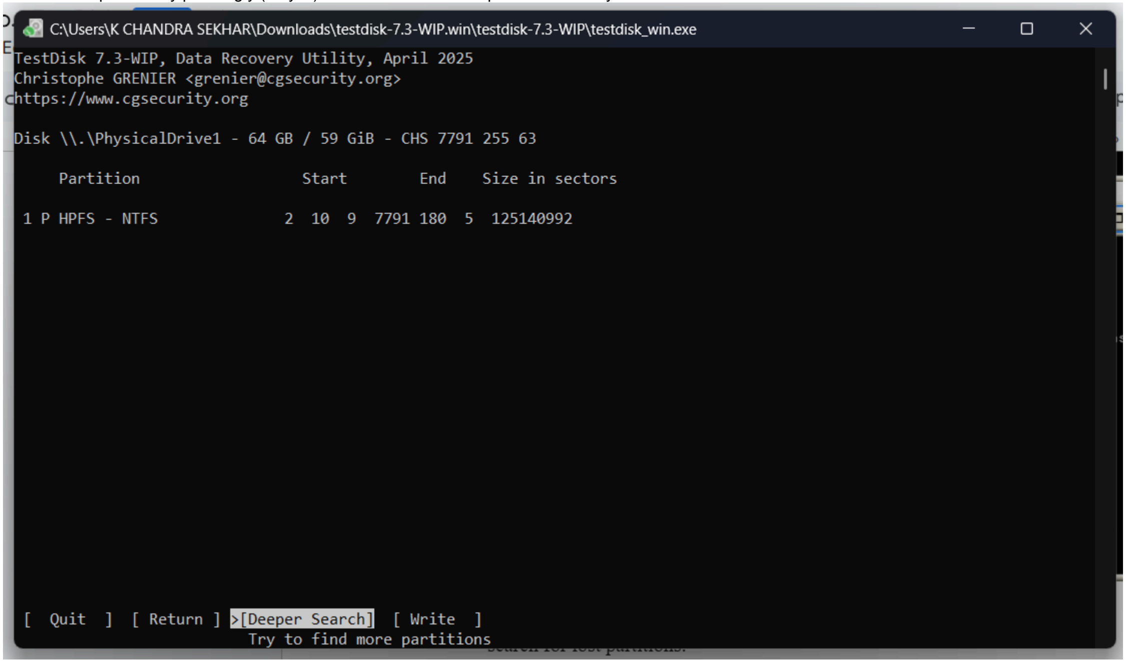


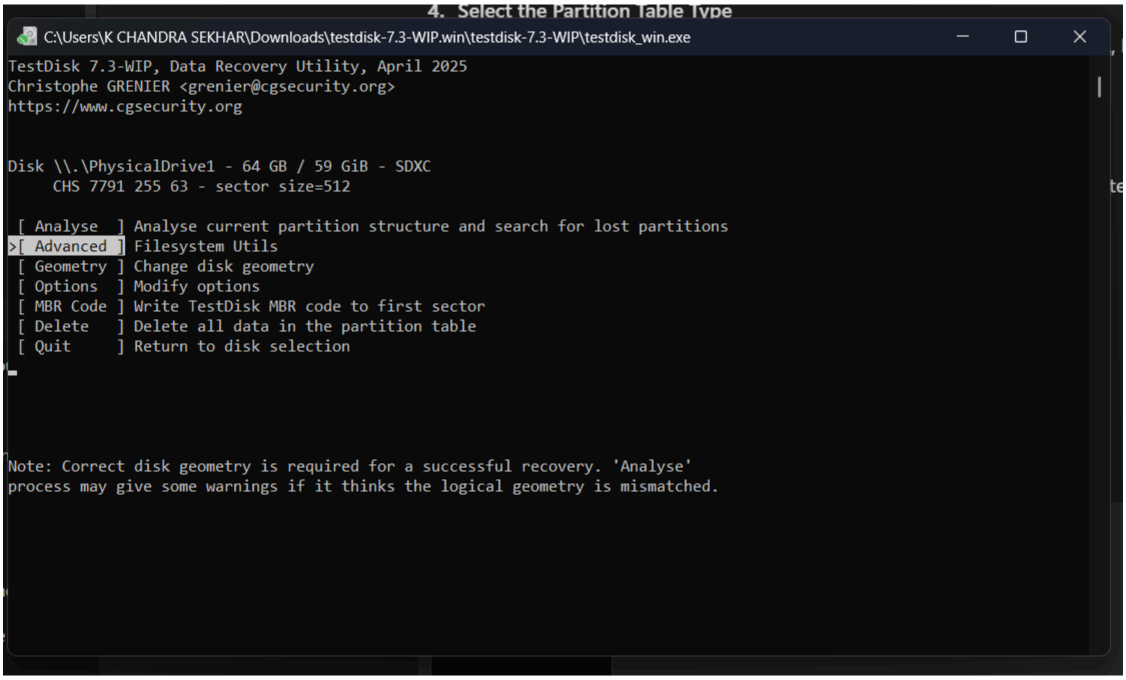
- Use Left/Right arrow keys to change status:
  - P: Primary
  - *: Bootable
  - L: Logical
  - D: Deleted
  - ensure that the partitions you want to recover are marked as Primary or Logical (and not deleted).

8. Write the Partition Table

- Once you are confident the partition structure is correct, select [Write] from the menu.

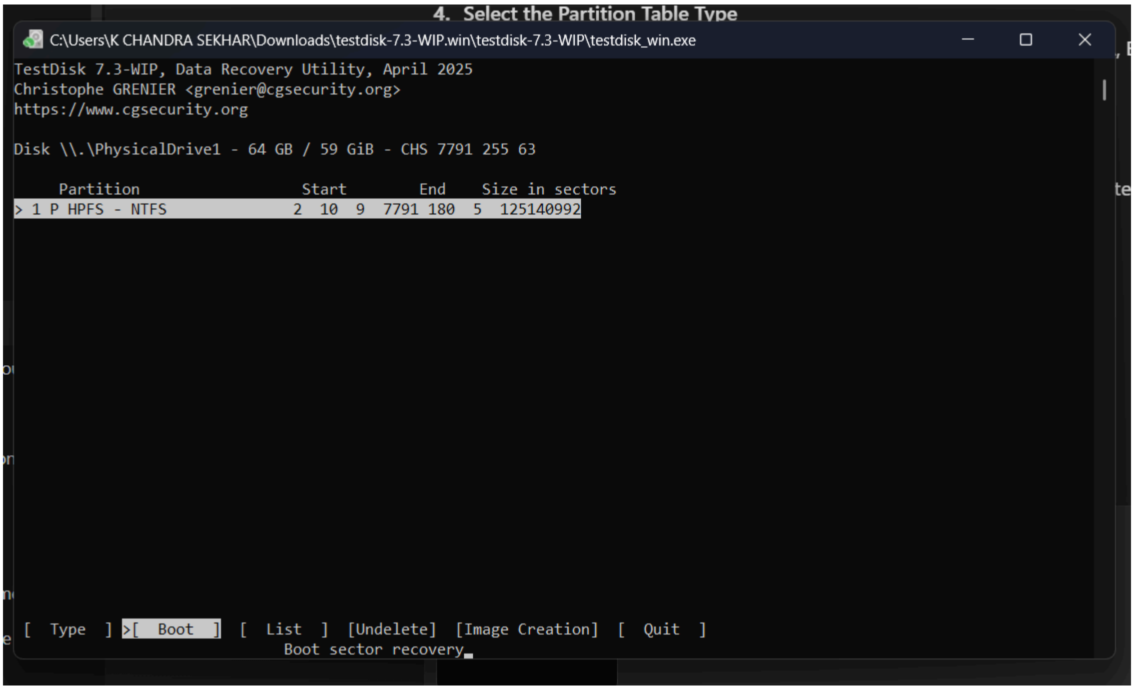
- Confirm the operation by pressing y (for yes). This will write the new partition table to your disk.



- 

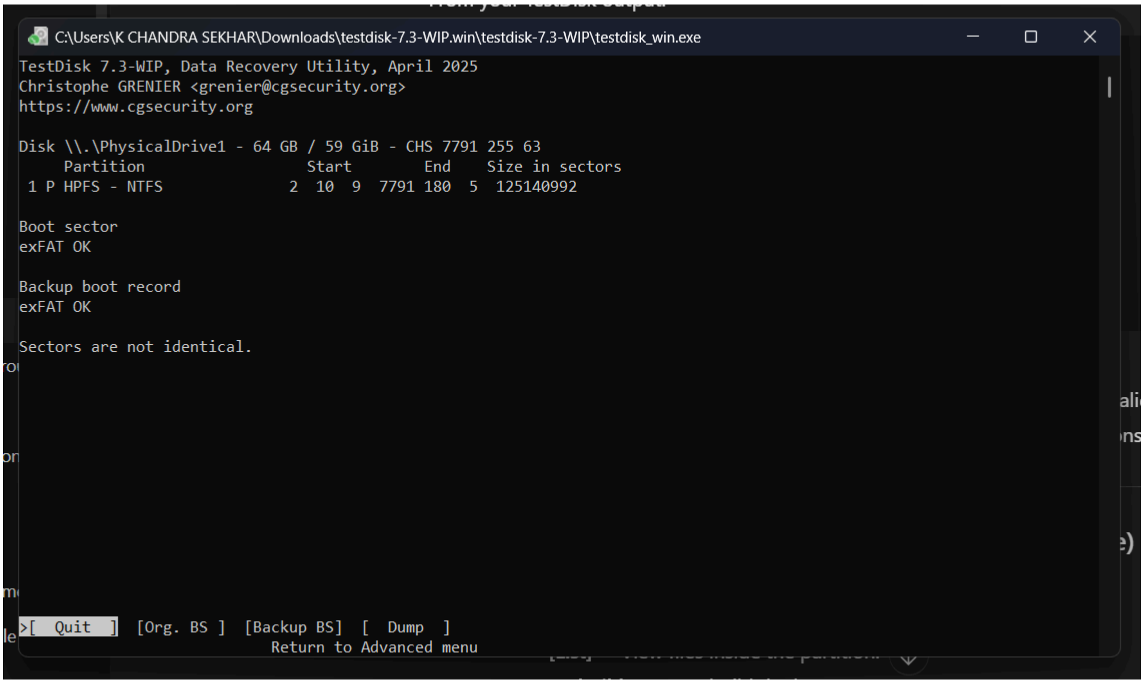
9. Recover Files

- If you just need to recover a few files without fixing the partition table, you can do so from the file list (after pressing p).
- Navigate to the folder containing your desired files.
- Use the colon : key to select the files you want to recover.
- Press the uppercase C key to copy the selected file(s).
- Navigate to a safe destination on a different storage device and press uppercase C again to paste.

- 

10. Exit and Restart

- Once your task is complete, select [Quit] to exit the program.
- If you wrote a new partition table to the drive, it is recommended to restart your computer to allow the operating system to recognize the changes.

- 

References:

https://www.cgsecurity.org/wiki/TestDisk_Download

https://www.cgsecurity.org/testdisk_doc/